Name = Krish

Roll Number = 2401010177

LAB MANNUAL = 04

```sql
CREATE DATABASE LabSheet4_DB;
USE LabSheet4_DB;

CREATE TABLE Student (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT
);

CREATE TABLE Course (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(50)
);
```

Tasks

Part A: User Creation

1. Create a new user named user1 with a password.
2. Create another user user2 with limited access.

```sql
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password1';

CREATE USER 'user2'@'localhost' IDENTIFIED BY 'password2';
```

Part B: Granting Privileges

3. Grant **SELECT and INSERT** privileges on a table Student to user1.
4. Grant **ALL privileges** on a table Course to user2.
5. Provide **WITH GRANT OPTION** to user1 and observe its effect.

```sql
GRANT SELECT, INSERT ON LabSheet4_DB.Student TO 'user1'@'localhost';

GRANT ALL PRIVILEGES ON LabSheet4_DB.Course TO 'user2'@'localhost';

GRANT SELECT, INSERT ON LabSheet4_DB.Student TO 'user1'@'localhost' WITH GRANT OPTION;
```

Part C: Revoking Privileges

6. Revoke **INSERT permission** from user1 on the Student table.
7. Revoke all privileges from user2 on the Course table.

```
REVOKE INSERT ON LabSheet4_DB.Student FROM 'user1'@'localhost';

REVOKE ALL PRIVILEGES ON LabSheet4_DB.Course FROM 'user2'@'localhost';
```

Part D: User Deletion

8. Drop/Delete user user2 from the database.

```
DROP USER 'user2'@'localhost';
```

9. Attempt to login using deleted credentials and observe the result.

```
SELECT user, host FROM mysql.user WHERE user = 'user2';
```

	user	host

Part E: Additional Commands

10. Display all users in the system.

```
SELECT user, host FROM mysql.user;
```

	user	host
▶	departments_user	localhost
	mysql.infoschema	localhost
	mysql.session	localhost
	mysql.sys	localhost
	root	localhost
	user1	localhost

11. Show privileges assigned to a specific user.

```
SHOW GRANTS FOR 'user1'@'localhost';
```

	Grants for user1@localhost
▶	GRANT USAGE ON *.* TO `user1`@`localhost`
	GRANT SELECT ON `labsheet4_db`.`student` ...